# 5.1 Maximum and Minimum Values; Critical Numbers

Name _____ Period ____ Date _____

AP Calculus 5.1 · Student sheet

**I can…**

- tell absolute extrema from local extrema.
- find the critical numbers of a function.
- find the absolute maximum and minimum of a continuous function on a closed interval.

**Vocabulary** · fill in the missing word

**Absolute maximum** — $f(c) \geq f(x)$ for every x in the _____

**Local maximum** — $f(c) \geq f(x)$ for every x in some open interval _____ c

**Critical number** — a number c in the _____ where $f'(c) = 0$ or $f'(c)$ does not exist

**Extreme Value Theorem** — a _____ function on a closed interval [a, b] has an absolute maximum and an absolute minimum

## Part 1 · Hills and valleys

$f(x) = x^3 - 3x$ on $[-2, 3]$.

| Point | x | f(x) | Slope there |
|---|---|---|---|
| Top of a hill | | | |
| Bottom of a valley | | | |
| Highest point on $[-2, 3]$ | | | |
| Lowest point on $[-2, 3]$ | | | |

## Part 2 · Notes and examples (fill in each note, then work the example)

**Extreme Value Theorem:** if f is continuous on [a, b], then f has an absolute maximum and an absolute _____ on [a, b].

### Example 1

$f(x) = x^2$ on $[-1, 2]$. Find the absolute maximum and minimum values.

**Critical numbers:** local extrema can only occur where $f'(c) = 0$ or $f'(c)$ _____.

### Example 2

Find the critical numbers of $f(x) = x^3 - 6x^2 + 9x + 2$.

**Closed interval method:** evaluate f at the critical numbers and at the _____; the largest value is the absolute max, the smallest the absolute min.

**Example 3**

Find the absolute extrema of $f(x) = x^3 - 3x$ on $[-3, 3]$.

**Example 4**

Find the critical numbers of $f(x) = x^{2/3}$.

**Example 5**

Find the critical numbers of $f(x) = \sin x + \cos x$ on $(0, 2\pi)$.

## Part 3 · Partner practice

**Partner 1**

Find the critical numbers of $f(x) = x^3 - 12x$.

**Partner 2**

(AP® Practice) The absolute maximum value of $f(x) = x^3 - 3x^2 + 1$ on $[-1, 3]$ is

(A) 1 (B) 3 (C) −3 (D) 5